

Scalability & Future Plan

Date	30 September 2026
Team ID	6ab39579211fa4fd5d28901b
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited subject coverage	Currently optimized for general educational topics, may not handle highly specialized domains accurately.	Medium
2	Text-based interaction only	Does not support image, video, or audio inputs yet.	Low
3	No user authentication or data persistence	User history and personalized data are not stored for future use.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Designed for small to moderate number of users (academic use).	Optimize backend and deployment to handle higher concurrent users using scalable cloud infrastructure.
2	Data Storage	No persistent storage implemented.	Integrate database to store user history, preferences, and learning progress.
3	Performance	Suitable for current usage with acceptable response time.	Implement caching and optimized API calls to improve response time for larger usage.
4	Security	Basic API key management using .env file.	Implement user authentication, secure data storage, and role-based access control.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Add user authentication and personalized profiles	3–4 months	Save user history, personalized learning experience, and better user management.
Phase 2	Support for multimedia inputs (images, audio, video)	6–8 months	Enable learning using different content types and improve accessibility.
Phase 3	Expand subject coverage and multi-language support	8–12 months	Make EduGenie useful for a wider audience with diverse learning needs.
Phase 4	Deploy on cloud with scalability	12+ months	Handle large user base, ensure high availability, and support future feature enhancements.